

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

**Condominium Booking and Operations
Management System**

In Partial Fulfillment of the Requirements
for the Degree of Bachelor of Science in Information Technology

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CHAPTER 1

INTRODUCTION

1:1 Background of the Study

The integration of information technology into business operations will become a critical factor in improving organizational efficiency, data accuracy, and service quality. Businesses across various industries will adopt computerized systems to streamline daily operations, reduce human error, and improve data management. In the real estate and property management industry, digital transformation will play a vital role in managing bookings, financial transactions, tenant records, and maintenance scheduling. As competition will increase and customer expectations will continue to evolve, property management businesses will need to adopt automated systems that will enhance operational control and decision-making. The growing demand for condominium rentals in urban areas will increase the need for efficient property management solutions. Many customers, particularly students, young professionals, and small families, will prefer condominium units because of their accessibility, security, and convenience. However, managing multiple rental units, tenant inquiries, booking schedules, payment monitoring, and maintenance requests will become challenging when handled manually. Gales Urban Haven, operating under the business name One Spatial Iloilo located in Iloilo City, will provide condominium rental and property management services to young professionals, students, and small families seeking affordable and convenient residential units. Despite its growing clientele, the business will currently rely on manual processes for recording bookings, tracking payments, computing revenue, and monitoring maintenance activities. Manual management of booking and operational records will present several risks, including data redundancy, record misplacement, delayed financial reporting, inaccurate revenue computation, and potential double bookings. The use of paper-based or spreadsheet-based systems will also limit accessibility, making it difficult for management to retrieve information quickly and generate accurate reports. Moreover, the absence of an integrated calendar and maintenance tracking system will limit the organization's ability to effectively monitor unit availability, schedule bookings, and address maintenance concerns promptly. These inefficiencies will affect both internal

operations and customer satisfaction. An automated booking and operations management system will address these challenges by centralizing business data into a single platform. Such systems will improve booking accuracy, simplify payment tracking, generate financial reports automatically, and provide better scheduling tools for administrators. Additionally, integrating maintenance management features will ensure that property concerns will be recorded, monitored, and resolved efficiently. This study will propose the development of a Condominium Booking and Operations Management System designed to automate booking records, payment tracking, revenue monitoring, calendar scheduling, and maintenance management through a centralized database-driven platform. The system will aim to improve operational efficiency, ensure data accuracy, reduce manual workload, and support informed decision-making for the business.

1.2 Statement of the Problem

Gales Urban Haven, operating under the business name One Spatial Iloilo, will currently manage condominium booking and operational processes using manual methods. These processes will include recording bookings, tracking tenant payments, computing revenue, scheduling reservations, and monitoring maintenance activities. While these methods may be manageable for a small number of clients, the increasing demand for condominium rentals will make manual operations inefficient, time-consuming, and prone to human error. The absence of an automated system will create operational challenges such as data redundancy, inaccurate record-keeping, delayed payment verification, revenue miscalculations, and double bookings. Additionally, the lack of an integrated scheduling and maintenance monitoring tool will affect the business's ability to properly manage unit availability and maintenance requests. These issues will reduce operational productivity and negatively impact customer satisfaction. This study will aim to develop a Condominium Booking and Operations Management System to address the identified operational concerns of the business.

Specifically, this study will seek to answer the following questions:

1. How will booking records be organized and managed efficiently to prevent data loss and double booking?
 2. How will payment tracking and revenue monitoring be automated to reduce computational errors and delays in financial reporting?
 3. How will an integrated calendar system improve scheduling and availability management?
 4. How will maintenance monitoring be systematically recorded and tracked to prevent operational conflicts?
 5. How will a centralized system enhance overall operational efficiency, accuracy, and decision-making?
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1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design, develop, and implement a centralized Condominium Booking and Operations Management System that will automate booking management, payment monitoring, revenue tracking, scheduling, and maintenance recording for Gales Urban Haven operating under the business name One Spatial Iloilo. The system is intended to improve operational efficiency, minimize manual errors, enhance data accessibility, and support faster and more accurate business decision-making.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. Analyze the existing booking and operational procedures of Gales Urban Haven to identify inefficiencies, operational challenges, and system requirements necessary for automation and process improvement.

2. Design a secure, accessible, and user-friendly web-based system that will provide administrators with an organized platform for managing condominium bookings, tenant information, payments, and maintenance records.
3. Develop an automated booking management module that will allow efficient recording of reservations, real-time unit availability checking, and prevention of scheduling conflicts and double bookings.
4. Implement a payment tracking system capable of recording tenant down payments, remaining balances, payment history, and total revenue to improve financial accuracy and monitoring.
5. Create a revenue reporting module that will automatically generate financial summaries, payment reports, and revenue analytics to reduce delays in financial reporting and support business planning.
6. Integrate a calendar-based scheduling system that will enable administrators to monitor reservations, tenant check-ins and check-outs, unit availability, and scheduled maintenance activities in a centralized calendar view.
7. Develop a maintenance monitoring module for recording, tracking, and managing repair requests, maintenance schedules, and service completion updates to ensure operational continuity and improved property upkeep.

8. Establish a centralized database management system that will securely store and organize booking records, payment transactions, tenant data, and maintenance logs for faster information retrieval and better record management.

9. Evaluate the functionality, usability, and efficiency of the developed system based on user feedback and system testing to determine its effectiveness in addressing the operational needs of the business.

1.4 Scope and Delimitation

This study will focus on the design, development, and evaluation of a web-based Condominium Booking and Operations Management System intended for use by Gales Urban Haven operating under the business name One Spatial Iloilo. The proposed system will be developed to automate and centralize major business operations related to condominium rental and property management.

The system will include the following core modules and functionalities:

- User Authentication Module for secure administrator login and access control.
- Booking Management Module for recording reservations, tenant information, booking history, and reservation details.
- Availability Checking Module for monitoring unit occupancy and preventing scheduling conflicts or double bookings.
- Payment Recording Module for documenting down payments, balance payments, payment history, and outstanding accounts.
- Revenue Monitoring and Financial Reporting Module for automated revenue computation, payment summaries, and generation of financial reports.
- Calendar Integration Module for scheduling reservations, tenant check-ins and check-outs, and monitoring unit availability.
- Maintenance Tracking Module for recording maintenance requests, monitoring repair schedules, and tracking service completion status.

The study will be limited only to the development, implementation, testing, and evaluation of the system within an academic setting. The project will be developed as a functional web-based prototype and will not cover large-scale commercial deployment.

Furthermore, the following features will not be included in the scope of the study:

- Integration with external online payment gateways or banking systems
- Mobile application development for Android or iOS platforms
- SMS or email notification services
- Multi-property or enterprise-level management features
- Cloud-based hosting and advanced cybersecurity implementations beyond standard authentication and database security

The evaluation of the system will be based on its functionality, usability, efficiency, reliability, and accuracy as assessed by selected evaluators.

1.5 Significance of the Study

The proposed Condominium Booking and Operations Management System is expected to provide benefits to various stakeholders by improving business processes and operational management.

- Business Owner-The study will help business owners improve operational efficiency through automated booking management, accurate payment monitoring, revenue tracking, and centralized record management. It will also support better decision-making through readily available reports and organized business data.

- **Administrative Staff**-The system will simplify daily operational tasks such as booking management, payment recording, schedule monitoring, and maintenance tracking. This will reduce manual workload, minimize errors, and improve productivity.
 - **Tenants and Clients**- Clients will benefit from improved booking organization, reduced scheduling conflicts, and more efficient handling of reservations and maintenance concerns, resulting in better service quality and customer satisfaction.
 - **BSIT Students**-This study will serve as a practical application of concepts in web development, database management, software engineering, and systems analysis and design. It will also provide students with experience in solving real-world business problems using information technology.
 - **Future Researchers**-The study may serve as a reference for future research related to property management systems, booking automation, revenue monitoring, and web-based business solutions. Researchers may use this study as a foundation for enhancing similar systems or exploring additional features and technologies.
 - **Academic Institution**-The institution may benefit from this study as an additional academic resource that demonstrates the practical use of information systems in solving business operational challenges and promoting innovation among students.
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1.6 Definition of Terms

The following terms are operationally defined to provide a clearer understanding of concepts used in this study:

- **Booking Management System** – A computerized system designed to record, organize, process, and monitor reservation transactions, including booking details, tenant information, unit availability, and reservation history. In this study, it refers to the

automated module used by One Spatial Iloilo for managing condominium bookings efficiently.

- **Condominium Booking and Operations Management System** – A web-based information system developed to automate booking processes, payment monitoring, revenue tracking, scheduling, and maintenance management for condominium rental operations. This serves as the proposed system in the study.

- **Revenue Monitoring** – The automated process of recording, tracking, and computing income generated from bookings, payments, down payments, and outstanding balances. In this study, it is used to improve financial accuracy and reporting efficiency.

- **Payment Tracking** – A system feature responsible for monitoring tenant payments, including initial down payments, remaining balances, payment history, and payment status. This ensures accurate financial record management.

- **Maintenance Tracking** – A system feature used to record, schedule, monitor, and update maintenance requests, repair activities, and service completion related to condominium units.

- **Calendar Integration** – A scheduling feature that visually organizes and displays booking schedules, tenant check-ins and check-outs, unit availability, and maintenance activities in a calendar format.

- **Centralized Database** – A structured digital repository used to securely store, organize, and manage system data such as booking records, tenant information, payment transactions, and maintenance logs in a single platform.

- **Automation** – The application of technology to perform repetitive operational processes with minimal manual intervention, reducing human error and improving efficiency.
 - **User Authentication** – A security mechanism that verifies the identity of authorized users through login credentials before granting access to the system.
 - **Web-Based System** – A software application accessible through a web browser using an internet or local network connection, allowing users to interact with the system online.
 - **Prototype** – A functional initial version of the proposed system developed for testing, evaluation, and demonstration purposes before full-scale implementation.
 - **Operational Efficiency** – The ability of an organization to perform business processes accurately, quickly, and with minimal resource wastage through optimized workflows and automation.
 - **Data Accuracy** – The correctness, reliability, and consistency of recorded information within the system, minimizing errors commonly associated with manual data handling.
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CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

According to Imkumner (2002), the proposed system handles most information management in this condominium such as customer information, room information, service information room renting, bills and receipts operation, and report for management team. Rental Condominium business is now becoming popular in the lodging market. Formerly the nature of the real-estate business, focus on the condominium market, was the construction methodology for selling. Halkiopoulou et al. (2020), the study focuses on exploration of knowledge for online booking systems and on the views of local students-users concerning the booking rate based on these online systems. Another perspective of this project is to investigate the decision-making process (emotion-focused) that they follow in order to choose a tourist destination via online booking systems.

Varutbangkul (2000), in Assumption University: Condominium management information system, this project is implemented to improve the efficiency of the existing system in Fair Tower, a condominium that belongs to GOS Limited Partnership, by designing a new computerized system to replace the old manual system in front office. Therefore, this project is aimed at providing a computerized system for the staff at the front desk in order to carry out their work more efficiently and accurately. The proposed system handles most information management in this condominium such as customer information, room information, service information room sale, room renting, bills and receipts operations, and reports for general manager.

According to Kuo et al. (2012), in Science Direct: Enhancement of condominium management based on the effect of quality attributes on satisfaction improvement (Expert Systems with Applications 39 (5), 5418-5425), condominium management (CM) businesses rely on continuous improvements to service quality (SQ) to increase consumer

satisfaction (CS) and thereby improve operations. Lacking knowledge of SQ and CS results in poor efficiency. To develop cost-effective strategies for improvements, the relationship between SQ and CS must be investigated. This work proposes novel procedures to explore the relationship between identified SQ attributes and CS in the CM business. First, we conduct exploratory factor analysis to cluster intrinsically similar attributes into three constructs and identify significant attributes for subsequent assessment. Furthermore, resident satisfaction scores (RSSs) are computed to assess the relative importance rate (RIR) of investigated attributes by applying an adapted multivariate method to regress the RSSs on the performance of SQ attributes. To facilitate the decision-making process, the service-quality model evolved from the integration of an improved Kano's approach and importance–performance analysis (IPA) is utilized to develop a prioritized strategy for satisfaction improvements in the CM business.

2.2 Local Studies

According to Bajado (2025), this research investigated the impact of hotel business tools and technologies on guest satisfaction in Tagaytay City, Philippines, a rapidly growing tourist destination. Hotels globally are adopting technology to enhance guest experiences and satisfaction. Gosela et al. (2024), it is a unified application aiming to address existing challenges by incorporating user-centric features. Through an advanced location-based services and streamlined booking processes, the application intends to enhance efficiency, accessibility, and sustainability in the tourism sector by providing real-time tracking, efficient transit solutions, and a comprehensive directory of establishments.

Gallenero et al. (2022), IEEE 14th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM), 1-5: Occupy-Real-Estate Management System with Sales Analysis and Customer Segmentation, the increase in population equates to the increasing demand of the real estate. Managing real estate manually takes a lot of time and

manpower, which is prone to damage and misplacement of documents and client's data. Customer support and community were also considered. Real-Estate Management System automates complex tasks and duties. Sales analysis through customer segmentation was implemented in the system to come up with better sales and understand the current, possible, and future customers of the real-estate company. Agile Methodology was applied in developing the system, in which different users – who has different roles and administrator rights, are needed to access, and manage the system. Testing, verification, validation, usability testing, and acceptance testing were done to ensure that the requirements needed were met, having both alpha and beta testing for the software.

2.3 Related Literature

The rapid advancement of information technology has significantly transformed the management practices of businesses in the hospitality, tourism, and real-estate industries. Traditional manual systems are gradually being replaced by computerized and web-based management systems to improve operational efficiency, customer satisfaction, accuracy of records, and decision-making processes. In condominium and rental management businesses, information systems play a vital role in organizing tenant records, monitoring room availability, handling payments, generating reports, and improving communication between management and clients.

According to Imkumnert (2002), computerized condominium systems help streamline the management of customer information, room details, service records, billing operations, and report generation. The study emphasized that rental condominium businesses require efficient information handling because manual processing often leads to delays, inaccuracies, and inefficient customer service. The integration of automated systems allows management teams to monitor transactions effectively and improve overall operational performance.

Similarly, Varutbangkul (2000) explained that condominium management information systems are essential in modern real-estate operations because they reduce paperwork, minimize human error, and improve transaction processing. The study highlighted the importance of replacing manual front-office operations with computerized systems to provide faster services, more accurate record keeping, and better monitoring of room sales, rentals, billing, and customer information. Through automation, employees can perform their tasks more efficiently while management gains better control over business operations.

In relation to service quality, Kuo et al. (2012) emphasized that customer satisfaction is strongly influenced by the quality of services provided by condominium management businesses. Their study revealed that service quality attributes such as responsiveness, reliability, convenience, and communication directly affect customer satisfaction levels. The researchers proposed the use of analytical models and service-quality evaluation methods to identify areas that require improvement. Their findings suggest that management systems should not only focus on operational efficiency but also on enhancing customer experiences and satisfaction.

The development of online booking technologies has also changed how customers interact with lodging and accommodation services. According to Halkiopoulou et al. (2020), online booking systems influence users' decision-making processes when selecting accommodations or tourist destinations. Their study focused on how digital booking platforms provide convenience, accessibility, and efficient information access to users. Online systems allow customers to compare options, check availability, and make reservations anytime and anywhere. This technological advancement highlights the growing demand for web-based systems in accommodation and property management industries.

In the Philippine context, technology adoption in hospitality and tourism businesses has become increasingly important. According to Bajado (2025), hotel business tools and technologies significantly affect guest satisfaction in tourism-driven locations such as Tagaytay City. The study found that digital systems improve customer experiences by

enabling faster transactions, convenient reservations, and more organized services. This demonstrates that customer-oriented technologies are becoming necessary for businesses that aim to remain competitive and efficient.

Furthermore, Gosela et al. (2024) discussed the importance of integrated applications with location-based services and streamlined booking features in improving tourism and transportation management. Their study emphasized that unified systems provide users with real-time information, accessibility, and efficient service delivery. The incorporation of booking and tracking functionalities enhances operational efficiency while promoting user convenience and sustainability in the tourism sector.

The increasing demand for efficient real-estate management systems was also highlighted by Gallenero et al. (2022). Their study on an Occupy-Real-Estate Management System explained that manual management processes consume excessive time and are prone to data loss and document misplacement. The researchers developed a computerized real-estate management system with sales analysis and customer segmentation features to improve business operations and customer support. The study further emphasized the importance of usability testing, validation, and agile development methodologies to ensure that the system effectively meets organizational requirements.

Overall, the reviewed literature and studies reveal that computerized management systems provide significant advantages in the areas of operational efficiency, customer satisfaction, data management, booking processes, and decision-making. These studies support the development of a modern condominium or rental management system that integrates automation, online services, and customer-centered functionalities. The existing literature also demonstrates the growing need for digital transformation in property management and hospitality industries to address the limitations of traditional manual processes.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local literature collectively emphasizes the growing importance of computerized and web-based management systems in the condominium, real-estate, hospitality, and tourism industries. The studies reveal that traditional manual processes are no longer sufficient in handling the increasing demands for efficient operations, accurate record management, faster transactions, and improved customer service.

Foreign studies highlighted the importance of automation in condominium management. Imkumnert (2002) and Varutbangkul (2000) both emphasized that computerized systems improve the efficiency of front-office operations by managing customer records, room information, billing, rentals, and reports more accurately and effectively. Their findings showed that replacing manual systems with automated solutions minimizes human errors, reduces paperwork, and improves operational performance.

Meanwhile, Kuo et al. (2012) focused on the relationship between service quality and customer satisfaction. Their study demonstrated that effective condominium management systems contribute to higher customer satisfaction through reliable and responsive services. Similarly, Halkiopoulos et al. (2020) stressed the importance of online booking systems in influencing customer decisions and improving convenience, accessibility, and user experience in accommodation services.

On the local level, the reviewed studies also support the integration of technology in property and hospitality management. Bajado (2025) found that the use of hotel technologies significantly improves guest satisfaction by providing faster and more organized services. In addition, Gosela et al. (2024) highlighted the effectiveness of integrated applications with booking and location-based services in enhancing operational efficiency and accessibility. Furthermore, Gallenero et al. (2022) emphasized that automated real-estate management systems help reduce workload, prevent data loss, improve customer support, and provide better business analysis.

Overall, the reviewed literature and studies strongly support the development of a computerized condominium or rental management system. The studies consistently show that automation improves operational efficiency, data accuracy, customer satisfaction,

and decision-making processes. They also reveal that modern users prefer systems that offer convenience, accessibility, and real-time services. Therefore, the proposed system is relevant because it addresses the common limitations of manual management processes and aligns with the current technological trends in property and accommodation management industries.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study will utilize a developmental research design, which will focus on the systematic planning, design, development, implementation, and evaluation of the proposed application. Developmental research design will be appropriate for the study because it will allow the researchers to create a functional and effective system intended to address existing problems encountered in manual condominium and rental management operations. The study will concentrate not only on the development of the application itself but also on evaluating its functionality, usability, reliability, and effectiveness in improving current business processes.

The proposed system will be developed to address common issues experienced in manual operations such as double bookings, misplaced records, delayed transactions, inaccurate computations, data redundancy, and inefficient report generation. Traditional methods of managing tenant information, room availability, reservations, billing, and payments often consume significant time and effort and may result in human errors that negatively affect customer satisfaction and operational efficiency. Through the developmental research approach, the researchers will identify these operational problems, analyze the needs of the users, and develop an automated solution that will improve the overall management process. The study will involve several stages including requirement gathering, system analysis, system design, coding and development, testing, evaluation, and implementation. During the requirement gathering phase, the researchers will collect relevant information regarding the current management practices, challenges, and user needs from the target users or stakeholders. The gathered data will serve as the foundation in designing the features and functionalities of the proposed application.

In the system design and development phase, the researchers will create the structure, interface, database, and functionalities of the application. The system will be designed to manage customer information, room reservations, booking schedules, billing transactions,

payment monitoring, and report generation efficiently and accurately. The application may also incorporate user authentication and security features to ensure the confidentiality and integrity of stored data.

After the development process, the system will undergo testing and evaluation to determine whether it meets the required standards and objectives of the study. Different testing procedures such as functionality testing, usability testing, reliability testing, and user acceptance testing will be conducted to identify system errors, evaluate performance, and ensure that the application operates according to user requirements. Feedback from respondents or evaluators will also be considered for possible improvements and refinements of the system.

The developmental research design will help ensure that the proposed application will be practical, user-friendly, and responsive to the needs of condominium or rental management businesses. Furthermore, this approach will support the study's goal of providing a modernized and efficient management solution that will minimize manual errors, improve operational efficiency, enhance customer service, and contribute to the digital transformation of property management operations.

3.2 System Development Methodology

The development follows the Agile Development Model with iterative phases:

- 1.Requirements Analysis – Analyze manual processes and identify system requirements.
- 2.System Design – Create a secure and user-friendly web interface.
- 3.Development – Build modules for booking, payment tracking, and maintenance.
- 4.Testing – Perform unit and system testing for reliability.
- 5.Deployment – Present the system as a functional prototype for evaluation.

3.3 System Architecture

The system follows a client-server architecture. Administrative staff interact with the web interface to manage bookings, payments, and maintenance requests while the server processes requests and stores data in a centralized MySQL database.

3.4 Tools and Technologies Used

- **Frontend:** React, TypeScript, Tailwind CSS , Vite
 - **Backend:** Node.js with Express.js Framework, RESTful API, JWT
 - **Database:** MySQL
 - **Development Environment:** Visual Studio Code
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3.5 Data Gathering Techniques

- **Process Analysis** – Identify inefficiencies in manual booking and payment tracking.
 - **Review of Existing Platforms** – Analyze property management systems to identify best practices.
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3.6 Testing Procedures

- **Unit Testing** – Test individual modules such as login and payment recording.
 - **System Testing** – Ensure integrated modules work correctly.
 - **User Acceptance Testing** – Verify the system meets operational requirements.
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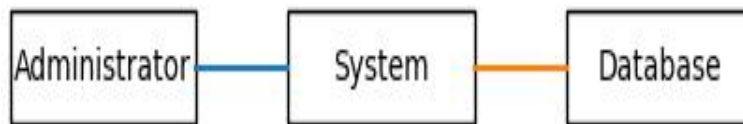
3.7 Evaluation Criteria

- **Functionality** – Accurate reservation recording.

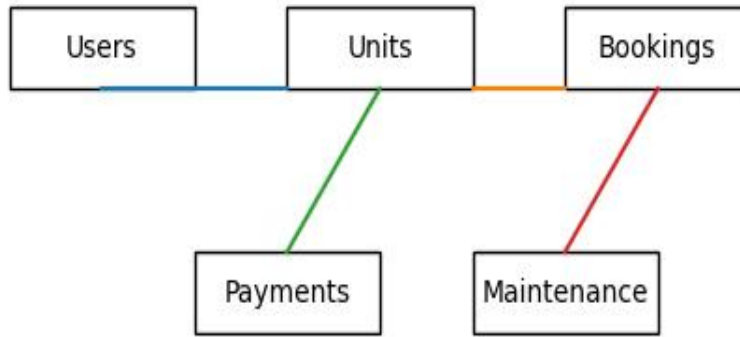
- **Usability** – Easy system navigation for staff.
 - **Accuracy** – Correct financial summaries.
 - **Security** – Authentication and database protection.
 - **Efficiency** – Reduced manual processes and prevention of booking conflicts.
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3.8 Supporting Diagrams

Data Flow Diagram (DFD Level 0)



Entity Relationship Diagram (ERD)



Use Case Diagram

